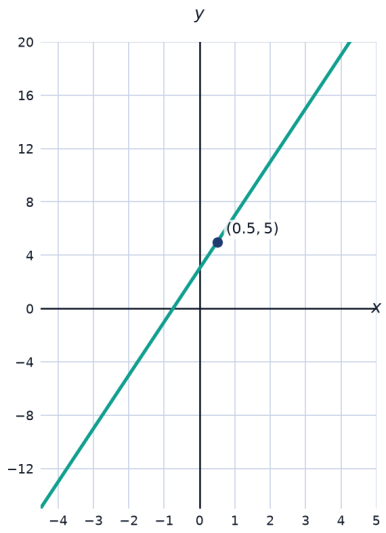
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

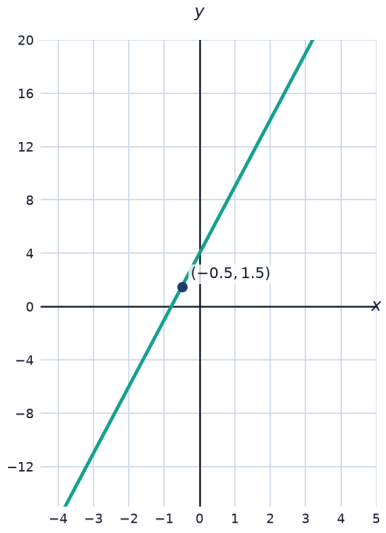
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(2)$.



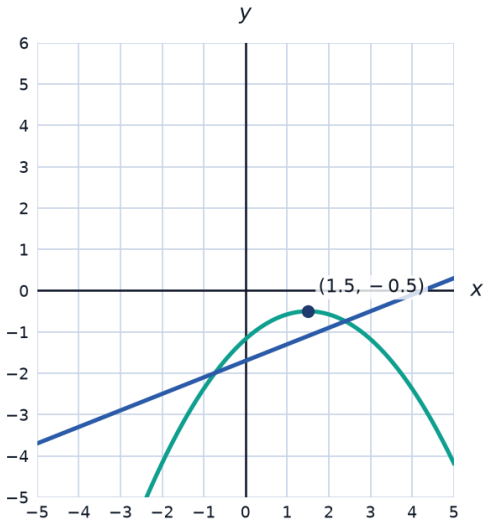
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for problem solving with trigonometric functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

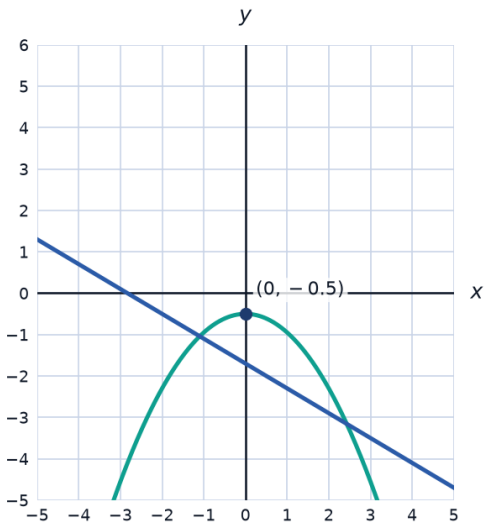


5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\, dx$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A weighted network has edges: DE = 1, BC = 2, AB = 3, CD = 4, BD = 5, CE = 6, AC = 7, AD = 8. Use Kruskal's algorithm to determine the minimum spanning tree and compute its total weight.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A network has weighted adjacency matrix W. Explain how the minimum spanning tree algorithm (Kruskal's) differs from Dijkstra's algorithm.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

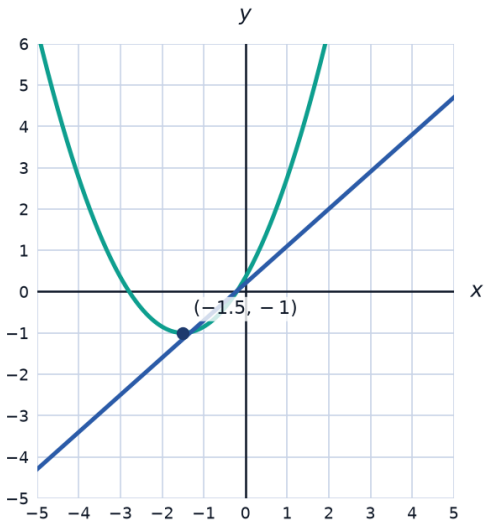
Find the Maclaurin series for $\ln(1+2x)$ up to the x^3 term, and state its interval of convergence.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[AA HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[AI HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all complex solutions to $z^4+16=0$ in exact rectangular form $a+bi$ using De Moivre's roots theorem.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3-3x=0$ on $x\in[-1,1]$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

(Paper 3 Extended Investigation) Define the integral family $I_n=\int_0^1 x^n e^x\,dx$ for integers $n\geq 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n\geq 1$. (c) Hence determine the exact value of $\int_0^1 x^3e^x\,dx$. (d) Prove that $0<I_n<\frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the rational function $f(x)=\frac{x^2-x-2}{x-3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

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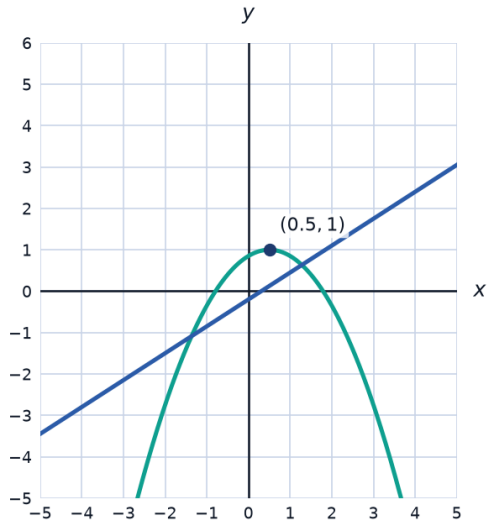
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Calculus

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Number & Algebra

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Functions

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Geometry & Trigonometry

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \le \pi$. Show that the sum of these five roots equals 0.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Statistics & Probability

In a circle of radius $10\ \text{cm}$, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

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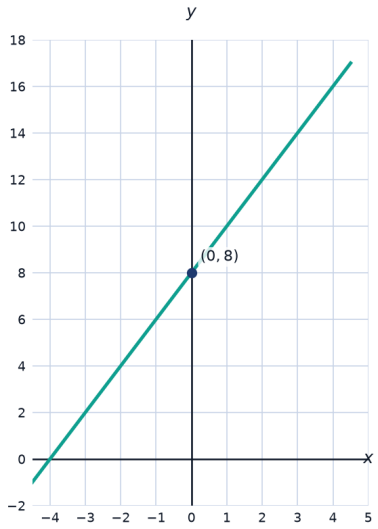
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Calculus

Prove by mathematical induction that $5^n - 1$ is divisible by 4 for all positive integers $n \in \mathbb{Z}^+$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(2)$.



a. 12

b. 14

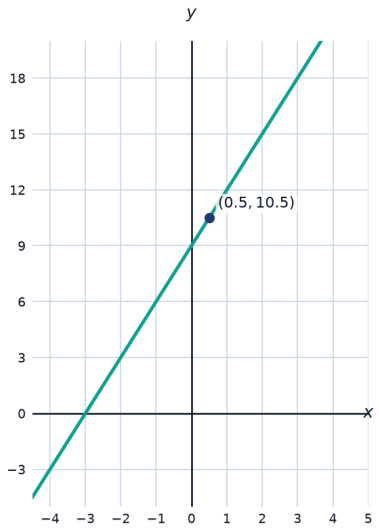
c. 4

d. 16

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(2)$.



a. 15

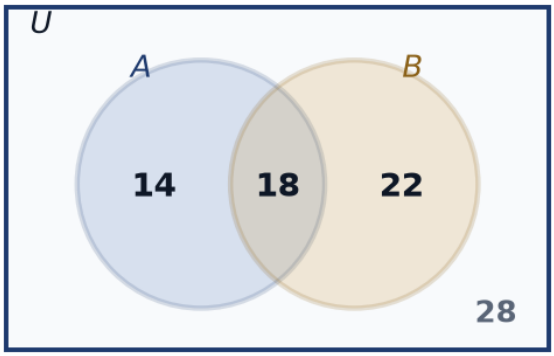
b. 18

c. 6

d. 19

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for amplitude, vertical shifts and dilations of trigonometric graphs consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?



a. 18.5

b. 19.5

c. 20.5

d. 22.5

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solutions to trigonometric equations consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 5$ in linear regression line, evaluate $f(3)$.

a. 20

b. 25

c. 15

d. 24

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

$x2500isinvestedatanannualrateof8\%$, compounded monthly, for 3 years. Find the value of the investment, correct to 2 decimal places.

a. 3175.59

b. 3145.59

c. 3115.59

d. 3205.59

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

$x2500isinvestedatanannualrateof6\%$, compounded quarterly, for 2 years. Find the value of the investment, correct to 2 decimal places.

a. 2756.23

b. 2846.23

c. 2816.23

d. 2786.23

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

$x2000isinvestedatanannualrateof6\%$, compounded semi-annually, for 2 years. Find the value of the investment, correct to 2 decimal places.

a. 2191.02

b. 2251.02

c. 2281.02

d. 2221.02

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A car purchased for $x20000depreciatesatarateof20\%$ per year (reducing balance). Find its value after 3 years, correct to 2 decimal places.

a. 10240.00

b. 9640.00

c. 10540.00

d. 9940.00

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

$x1500isinvestedatacompoundinterestrateof7\%$ per year, compounded annually, for 5 years. Find the value of the investment, correct to 2 decimal places.

a. 2103.83

b. 2063.83

c. 2143.83

d. 2023.83

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Site $A(1, -1)$ and site $B(8, 1)$ generate a Voronoi diagram. Determine which site the point $P(5, -1)$ is closer to.

a. Site B

b. Site A

c. Equidistant from A and B

d. Cannot be determined

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

$x1200isinvestedatasimpleinterestrateof8\%$ per year for 5 years. Find the interest earned.

a. 480.00

b. 470.00

c. 490.00

d. 500.00

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Given $\log_5(6) = p$, express $\log_5(6^3)$ in terms of p.

a. $p + 3$

b. $3p$

c. $\frac{p}{3}$

d. p^3

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve for x, correct to 2 decimal places: $2^{x+2} = 41$.

a. 2.86

b. 2.36

c. 3.36

d. 3.86

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Convert $\frac{\pi}{2}$ radians to degrees.

a. 315°

b. 30°

c. 240°

d. 90°

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic sequence has first term $u_1 = 4$ and common difference $d = -4$. Find S_{12}, the sum of the first 12 terms.

a. -198

b. -240

c. -216

d. -221

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A bag contains 5 blue balls, 5 green balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{7}{12}$

b. $\frac{1}{2}$

c. $\frac{5}{12}$

d. $\frac{1}{6}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Geometry & Trigonometry

Use the change of base formula to evaluate $\log_2(13)$ correct to 2 decimal places.

a. 4.00

b. 3.70

c. 3.40

d. 3.10

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Statistics & Probability

A bag contains 6 yellow balls, 3 blue balls, and 5 red balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{3}{7}$

b. $\frac{9}{14}$

c. $\frac{3}{14}$

d. $\frac{5}{14}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Calculus

A bag contains 5 blue balls, 4 yellow balls, and 8 red balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{5}{17}$

b. $\frac{9}{17}$

c. $\frac{4}{17}$

d. $\frac{8}{17}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number & Algebra

A bag contains 8 blue balls, 6 red balls, and 5 yellow balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{5}{19}$

b. $\frac{6}{19}$

c. $\frac{8}{19}$

d. $\frac{13}{19}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Functions

A bag contains 4 green balls, 3 blue balls, and 2 red balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{4}{9}$

b. $\frac{2}{9}$

c. $\frac{1}{3}$

d. $\frac{2}{3}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Geometry & Trigonometry

A bag contains 4 red balls, 2 blue balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{1}{2}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{3}{4}$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Statistics & Probability

A bag contains 3 red balls, 3 green balls, and 2 blue balls. A ball is drawn at random. Find $P(\text{blue})$.

a. $\frac{3}{4}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{8}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Calculus

A bag contains 8 blue balls, 4 red balls, and 2 green balls. A ball is drawn at random. Find $P(\text{red})$.

a. $\frac{1}{7}$

b. $\frac{4}{7}$

c. $\frac{2}{7}$

d. $\frac{5}{7}$

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